

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456145.8852 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.045
Transit depth (Rp/Rs)^2	2.06 +/- 1.3 [%]
Orbital Inclination (inc)	83.79 +/- 2.85
Airmass coefficient 1 (a1)	1.22 +/- 0.59
Airmass coefficient 2 (a2)	-0.09 +/- 0.26
Scatter in the residuals of the lightcurve fit is	50.37 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	7.25
Transit Duration (day)	0.041 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.045 vs 0.1326 (deviation 0.19 σ)
Duration fit vs published	0.041 d vs 0.0737 d (deviation 0.97 σ)
Mid-time O-C	-6.3 min
Depth SNR	2.6
Residual scatter	50.37 %

Light curve

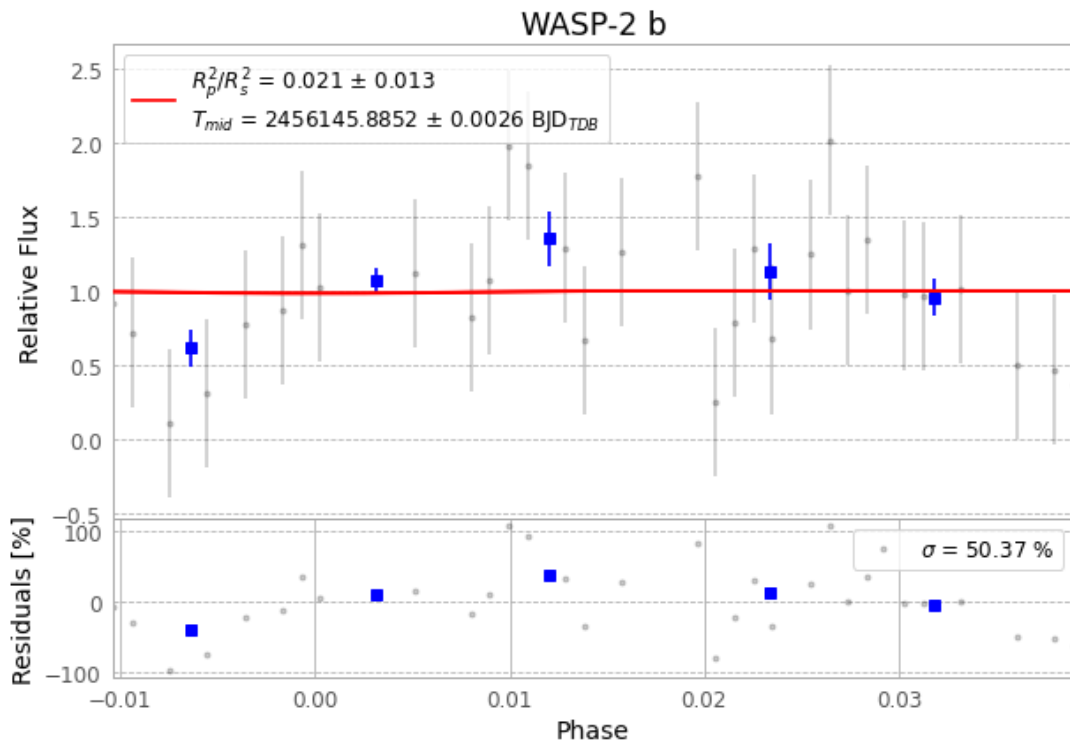


Figure 1 — FinalLightCurve_WASP-2 b_2012-08-06.png (SHA-256 5cdbd7860e8ae72f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [262, 372], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 65931126d1f06c06...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

56 FITS frames (37 MB), WASP-2120806083312.FITS → WASP-2120806112113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-120806.log (6caa87696673...), FinalLightCurve_WASP-2 b_2012-08-06.png (5cdbd7860e8a...), FinalLightCurve_WASP-2 b_2012-08-06.csv (c060a77f917e...)

Compute resources

Wall time 163.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:14Z.